

Introduction to Programming

Programming is a way to give instructions to a computer so it can perform a specific task. To start, we need to understand how to communicate with the computer in a language it can understand.

Key Points

- **Programming** is a process of giving instructions to a computer to perform a specific task.
- A **program** is a set of instructions that a computer can execute.
- **Input** is the data that we provide to the computer, and **output** is the result that the computer gives us after processing the input.
- **Processing** involves the computer performing calculations or operations on the input data.

Understanding the Problem Statement

To write a program, we need to understand the **problem statement**, which is a clear description of what we want the computer to do. We also need to identify the **inputs** and **outputs** of the program.

Breaking Down the Problem

We can break down a problem into smaller steps:

1. **Read** the input data.
2. **Process** the input data to get the desired output.
3. **Print** the output.

Creating a Flowchart

A **flowchart** is a diagram that represents the steps involved in solving a problem. It helps us to visualize the flow of the program and ensure that we have not missed any steps.

Key Takeaways

- **Programming** involves giving instructions to a computer to perform a specific task.
- We need to understand the **problem statement**, **inputs**, and **outputs** to write a program.
- **Flowcharts** help us to visualize the steps involved in solving a problem and ensure that we have not missed any steps.
- **Input**, **processing**, and **output** are the basic components of a program.